# Class 13 Quiz

Name: _____

**Notation.** $L$ = labor, $K$ = capital, $Q$ = output, $W$ = wage (price of labor), $R$ = rental rate (price of capital).

$$MPL = \frac{\Delta Q}{\Delta L}, \qquad MPK = \frac{\Delta Q}{\Delta K}.$$

The marginal product of labor ($MPL$) is the extra output from one more unit of labor, holding capital fixed. The marginal product of capital ($MPK$) is the extra output from one more unit of capital, holding labor fixed.

**1.** In words, define the marginal rate of technical substitution of labor for capital, $MRTS_{LK}$. What does it measure? What would $MRTS_{LK} = 2$ mean for the firm?

**Use the following.** The marginal rate of technical substitution equals the ratio of marginal products, $MRTS_{LK} = \dfrac{MPL}{MPK}$. A firm minimizes the cost of producing a given output where the isoquant is tangent to the lowest isocost line:

$$\underbrace{-\frac{W}{R}}_{\text{slope of isocost}} = \underbrace{-\frac{MPL}{MPK}}_{\text{slope of isoquant}} \quad \Longrightarrow \quad \frac{MPL}{W} = \frac{MPK}{R}.$$

**2.** Suppose that at the firm's current input mix, $\dfrac{MPK}{R} > \dfrac{MPL}{W}$. Interpret this inequality. Is the firm minimizing cost? Should it use more capital or more labor? Explain.

**3.** Suppose instead that $\dfrac{MPL}{W} > \dfrac{MPK}{R}$. Interpret this inequality. Should the firm use more capital or more labor? Explain.

# Class 13 Activity

## Problem 1 (7.14)

You are the CEO of a major producer of funnel cakes. Your cost accountant has provided you with a table describing your cost structure, but you have inadvertently dripped cooking grease on it and most of the table is illegible. Reconstruct the table below, given the remaining legible numbers.

| $Q$ | $TC$ | $FC$ | $VC$ | $MC$ | $AVC$ | $AFC$ | $ATC$ |
|---|---|---|---|---|---|---|---|
| 0 | | | | – | – | – | – |
| 1 | | | 17 | | | | |
| 2 | | | | 15 | | | |
| 3 | 101 | | | | | | |
| 4 | | | | | 14.5 | | |
| 5 | 122 | | 67 | | | 11 | |
| 6 | | | | | | | 21 |

## Problem 2 (7.16)

Suppose a firm has the following production function: $Q = 2KL$. The marginal product of capital for this production function is $2L$, and the marginal product of labor is $2K$. If capital rents for \$100 per unit per day, labor can be hired for \$200 per unit per day, and the firm is minimizing costs,

**a.** Find the total cost of producing 60 units of output.

**b.** Suppose you want to find the cost of producing a more general $q$ units of output. Using the same set of procedures you followed in (a), but using a more general $q$ instead of 60, find this firm's total cost function, which expresses total cost as a function of $q$.

**c.** Divide the total cost function you found in (b) by $q$ to find this firm's average cost function.

## Problem 3 (8.8)

Hack's Berries faces a short-run total cost of production given by

$$TC = Q^3 - 12Q^2 + 100Q + 1{,}000,$$

where $Q$ is the number of crates of berries produced per day. Hack's marginal cost of producing berries is $MC = 3Q^2 - 24Q + 100$.

**a.** What is the level of Hack's fixed cost?

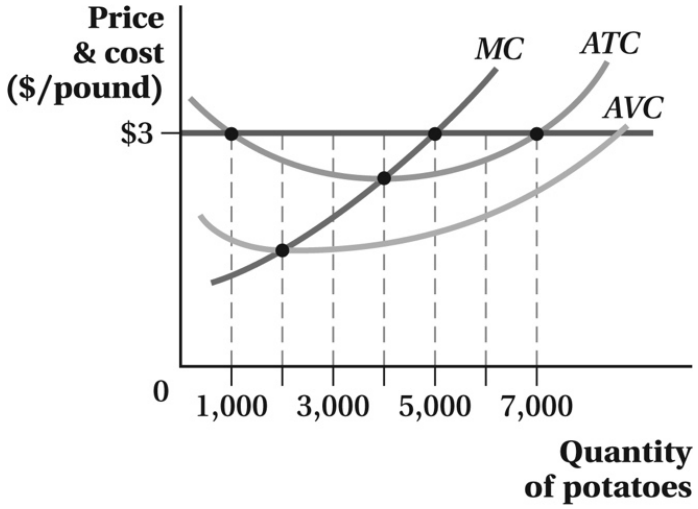
**b.** What is Hack's short-run average variable cost of producing berries?

**c.** If berries sell for \$60 per crate, how many berries should Hack produce? How do you know? (*Hint:* You may want to remember the relationship between $MC$ and $AVC$ when $AVC$ is at its minimum.)

**d.** If the price of berries is \$73 per crate, how many berries should Hack produce? Explain.

## Problem 4 (8.12)

Consider the graph below, which depicts the cost curves of a perfectly competitive seller of potatoes. Potatoes currently sell for $3 per pound.




**a.** To maximize profit, how many pounds of potatoes should this seller produce?

Suppose that the potato grower's bank ratchets up the interest rate applicable to the grower's adjustable-rate mortgage loan. This increases the size of the potato grower's monthly mortgage payment.

**b.** Illustrate the change in the mortgage payment by shifting the appropriate cost curves.

**c.** Which curves shift? Which do not? Why?

**d.** How does the change in interest rates affect the grower's decision on how many potatoes to produce?

**e.** What happens to the potato grower's profit as a result of the increased interest rate?

**f.** How does the change in interest rates affect the shape and/or position of the grower's short-run supply curve?